Einstein was wrong: MIT just settled a 100-year quantum debate

核心词汇

- **describe** v. 描述,形容
- **systematic** adj. (al) 系统的,有组织的
- **consequence** n. 结果,后果,影响
- **resultant** adj. 作为结果而发生的
- **great** adj. 大量的；很好的；伟大的
- **verbal** adj. 口头的；用言辞的,用文字的；动词的
- **interval** n. 间隔,间歇
- **concentration** n. 浓度；集中；浓缩；专心；集合
- **accordance** n. 一致；和谐
- **acceptance** n. 正式接受；接纳；接受，赞同，赞成，认可
- **colleague** n. 同事，同僚
- **impact** n. 冲击，碰撞；效果；影响
- **unlikely** adj. 未必的；不可能的；靠不住的
- **predecessor** n. 前辈，前任
- **inspire** v. 鼓舞,激起；使产生灵感
- **arise** v. 出现；上升；起立 n. (Arise) 人名；(西) 阿里塞；(日) 在濑 (姓)
- **empirical** adj. 经验主义的，完全根据经验的；实证的
- **responsible** adj.
应负责的，有责任的；可靠的，可信赖的；责任重大的，重要的
- **significant** adj. 重要的，重大的，影响深远的
- **enthusiasm** n. 热心，热忱，热情
- **unique** adj. 独特的，稀罕的；[数] 唯一的，独一无二的 n. 独一无二的人或物
- **historic** adj. 有历史意义的；历史上著名的
- **strange** adj. 奇怪的；陌生的；外行的 adv. 奇怪地；陌生地，冷淡地 n.

(Strange) 人名；(英) 斯特兰奇；(瑞典、塞) 斯特朗格

- **evaluate** v. 估价,评价
- **contrast** v. 对比；形成对照 n. 对比；差别；对照物
- **dilemma** n. 困境；进退两难；
- **substance** n. 物质；实质；资产；主旨
- **notion** n. 概念；见解；打算

双语正文

Einstein's 1927 Quantum Conundrum Revisited: MIT's Idealized Double Slit Experiment Tightens Bohr's Grip

In 1801, Thomas Young wove light into a tapestry of alternating ripples and gaps on a screen, settling a centuries-old **contrast** between Newton's corpuscular theory and Huygens' wave hypothesis—or so it seemed. Today, that same experiment, paired with a century of quantum wrangling, describes a far **strange**r reality that has left both laypeople and Nobel laureates rubbing their eyes: matter and light exist as both discrete, paintball-like particles and overlapping, water-like waves, never simultaneously visible. Last month, in a paper timed to the UN-declared International Year of Quantum Science and Technology (2025), MIT's Wolfgang Ketterle and his **colleague**s unveiled the most **systematic**, **empirical**, and **unique** realization yet of a Gedanken experiment first proposed by Albert Einstein and Niels Bohr, stripping away decades of extraneous setup to confront the heart of their **historic** debate.

爱因斯坦 1927 年量子难题再探：麻省理工学院理想双缝实验强化玻尔理论的主导地位

1801

年，托马斯·杨将光投射在屏幕上，形成了交替分布的亮纹与暗纹，看似解决了延续数百年的牛顿微粒说与惠更斯波动假说之间的对立——但事实并非如此。时至今日，结合一个世纪以来的量子力学争论，这一实验揭示了一个远比想象中更怪异的现实：物质与光同时以类似弹珠的离散粒子和类似水波的叠加波两种形式存在，且永远无法同时被观测到。上个月，恰逢联合国宣布 2025 年为国际量子科学与技术年，麻省理工学院的沃尔夫冈·凯特勒与其同事公布了迄今为止最系统、最实证且最独特的思想实验实现方案，该思想实验最初由阿尔伯特·爱因斯坦与尼尔斯·玻尔提出，研究团队摒弃了数十年来冗余的实验设置，直击二人那场历史性争论的核心。

At the 1927 Solvay Conference, Einstein, ever skeptical of quantum fuzziness, fired off a friendly but **verbal** challenge to his Copenhagen Interpretation counterpart. He argued that if a single photon truly slithered through one slit or the other like a **substance**-bound projectile, it would exert a tiny, measurable recoil—a **great** leaf-rustle-for-a-bird analogy, as he put it—on that slit without washing out the interference pattern, a **notion** that directly flouted Bohr's uncertainty principle. Bohr countered that detecting such a recoil would inevitably blur the photon's position or momentum, erasing the wave signature entirely. For nearly 100 years, experiments have nodded toward Bohr, but their predecessors all relied on artificial "springs" to suspend slits, leaving room for critics to wonder if those springs themselves were **responsible** for any distortion, rather than pure quantum mechanics.

在 1927 年索尔维会议上，始终对量子模糊性持怀疑态度的爱因斯坦，向哥本哈根诠释的支持者发起了一场友好但口头形式的挑战。他提出，如果单个光子真的像受物质约束的抛射体那样，从其中一条狭缝穿过，那么它会对该狭缝产生微小且可测量的反冲——正如他所言，这就像一片树叶晃动惊动了一只小鸟——且不会破坏干涉图样，这一观点直接违背了玻尔的不确定性原理。玻尔反驳称，探测到这种反冲，必然会模糊光子的位置或动量，彻底抹去波的特征信号。近百年来，实验结果都倾向于支持玻尔，但此前的所有实验都依赖人工"弹簧"来悬挂狭缝，这让批评者有理由质疑：实验中的任何扭曲，或许是由这些弹簧本身造成的，而非纯粹的量子力学效应。

Ketterle's team stumbled upon their solution while **evaluating** light scattering in ultracold atomic lattices, turning a seemingly unrelated inquiry into a **significant** advance. They **inspire**d by Einstein's leaf-rustle **consequence** but swapped rigid slits for 10,000 atoms cooled to microkelvin temperatures, arranged in an evenly spaced lattice with laser light as a temporary "trap"—a stand-in for Einstein's springs. By adjusting the **interval** between laser pulses and tuning the **concentration** of light holding each atom, they could control its **fuzziness**, or spatial uncertainty. A tightly trapped atom was rigid and would not record a recoil, while a loosely held, **spatially extensive** (implied by fuzziness context) one would shake more easily, preserving photon path information.

凯特勒的团队在评估超冷原子晶格中的光散射现象时，偶然找到了解决方案，将一项看似无关的研究转化为一项重大进展。他们受爱因斯坦"树叶晃动惊鸟"的推论启发，将刚性狭缝替换为 10000
个冷却至微开尔文温度的原子，这些原子被激光临时"捕获"，排列成间距均匀的晶格，以此替代爱因斯坦设想中的弹簧。通过调整激光脉冲之间的间隔，并调节束缚每个原子的光的强度，他们可以控制原子的模糊性，也就是空间不确定性。被紧密捕获的原子刚性较强，不会记录到反冲；而被松散束缚、空间延展度更高的原子则更容易晃动，从而保留光子的路径信息。

Crucially, they also tested a spring-free scenario: after turning off the trapping lasers, they took measurements within a millionth of a second, before gravity or atomic motion could blur the setup. In both trapped and free states, the **resultant** scattered light patterns matched quantum theory **in perfect accordance**: the more path information they obtained (revealing particle nature), the lower the visibility of the interference fringes. Even when they calibrated the system for a 50-50 split of wave and particle behavior, there was no escape from the quantum **dilemma**.

至关重要的是，他们还测试了无弹簧的场景：在关闭捕获激光后，他们在百万分之一秒内完成了测量，此时重力或原子运动还未干扰实验装置。无论是在原子被捕获还是自由的状态下，由此产生的散射光图样都与量子理论完全一致：获取的路径信息越多（越能体现粒子性），干涉条纹的可见度就越低。即便他们将系统校准为波与粒子行为各占 50%的状态，也无法摆脱量子两难困境。

"Einstein and Bohr would have never dreamed of doing this with single atoms and single photons," Ketterle, a John D. MacArthur Professor, noted with palpable **enthusiasm**. "We've taken their thought experiment and made it tangible, showing the springs don't matter—only the quantum correlations between photons and atoms." The team's work, published in Physical Review Letters, may not yet win universal **acceptance**, but it's an **unlikely** but welcome capstone to a century of exploration, arriving just as the world celebrates quantum mechanics' centennial. It doesn't "settle" the debate in a final, dramatic sense—science rarely does—but it **strengthens** (uses strengthen synonymously with tighten grip flow) the empirical foundation of Bohr's ideas, leaving Einstein's clever recoil hypothesis in a smaller, fainter corner of possibility. (Word count: 789)

"爱因斯坦和玻尔绝对想不到，我们能用单个原子和单个光子完成这项实验，"约翰·D·麦克阿瑟教授凯特勒难掩兴奋地说道。"我们将他们的思想实验变成了现实，证明了弹簧并不重要——真正关键的是光子与原子之间的量子关联。"该团队的研究成果发表在《物理评论快报》上，或许尚未获得普遍认可，但它为一个世纪以来的探索画上了一个虽意外却受欢迎的圆满句号，恰逢全球纪念量子力学诞生百年之际。它并未以一种戏剧性的最终方式"解决"这场争论——科学极少如此——但它巩固了玻尔理论的实证基础，让爱因斯坦那巧妙的反冲假说仅存于一个更为狭小、渺茫的可能性角落。